Pal et al. 2021 · Breast Cancer

10x Chromium · 59,359 malignant cells · 22 samples (ER+, HER2+, TNBC)

Question: do malignant cells recapitulate embryonic-development programs?

GO / NO-GO GATES

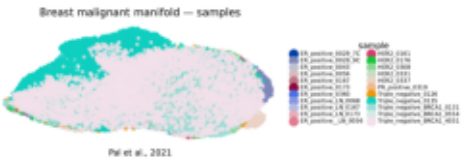
Gate 1 **GO** Batch mixing + continuity, judged on the regenerated embedding across a theta x n_pcs sweep (per-neighbourhood iLISI). At theta=10/n_pcs=20 mixing plateaus (median iLISI 4.96 of ~22); every UMAP band holds 18-19 samples on one connected manifold (8 island cells removed).

Gate 2 **GO** Module geography (non-branching). One organized developmental axis: pSom high at the root, Crest high at the terminus (Crest<->pSom spatial r=-0.93; Gast<->pSom r=-0.57), an inverse terminal Crest gradient -> GO. FLAG: Crest co-enriches the Gast side (r=+0.40 vs pSom -0.93), an inversion.

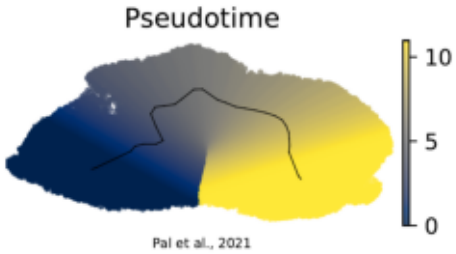
Gate 3 **NA** Branching-only. The manifold is non-branching (PPT forks scale with node count 10->21; a curve fits fold-free, 2 tips / 0 forks), so there is no bifurcation to test for Gast<->pSom cross-repression.

STEP FIGURES

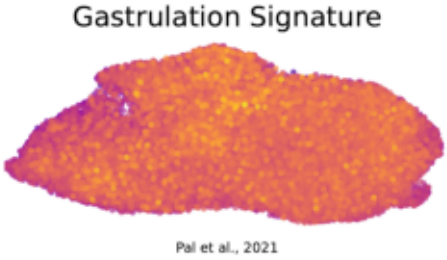
01 · Samples (Gate 1)



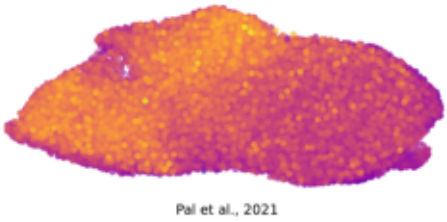
02 · Pseudotime



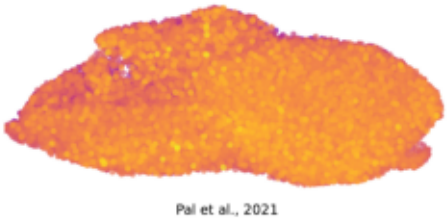
03 · Gastrulation



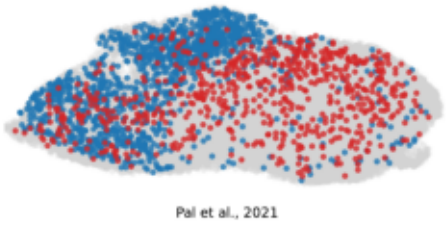
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



PARAMETER CHOICES

Sample threshold:
>=12 cells (geometric mean of the 56->3 gap); 22/26 samples kept

Integration:
Harmony on `sample`, theta=10 (iLISI plateau 5->10), n_pcs=20

Overdispersed genes:
3,733 (scFates find_overdispersed)

Island removal:
radius graph 6x median-NN, largest component (8 cells dropped)

Geometry:
Non-branching (ElPiGraph curve; 2 tips, 0 forks, fold-free s.a.=2.52)

Root:
pSom/Gast-high, Crest-low tip (origin; cyc rho=-0.57)

Module strengths (mean):
Gast -0.007 · pSom +0.036 · Crest -0.034 (Linear-pSom/Crest)

EMT-TFs:
PRRX2 (early/root) <-> SNAI1 (late/terminus)

VERDICT

The embryonic pattern reproduces as a non-branching axis (G1 GO / G2 GO / G3 NA): a single gastrulation/somitogenesis origin resolves toward a neural-crest terminus, with no Gast/pSom bifurcation. Crest-arm inversion flagged. Status: Gate 2 GO, non-branching.